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GEOG 450

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The Colors of a Seattle Sky

Introduction

This assignment asked us to find the dominant colors of a location in Seattle over several days. Our group chose to document the colors of the sky near Smith Hall on the main UW campus. This report covers our data collection process, the infographic we produced, and some thoughts on the project.

Procedures

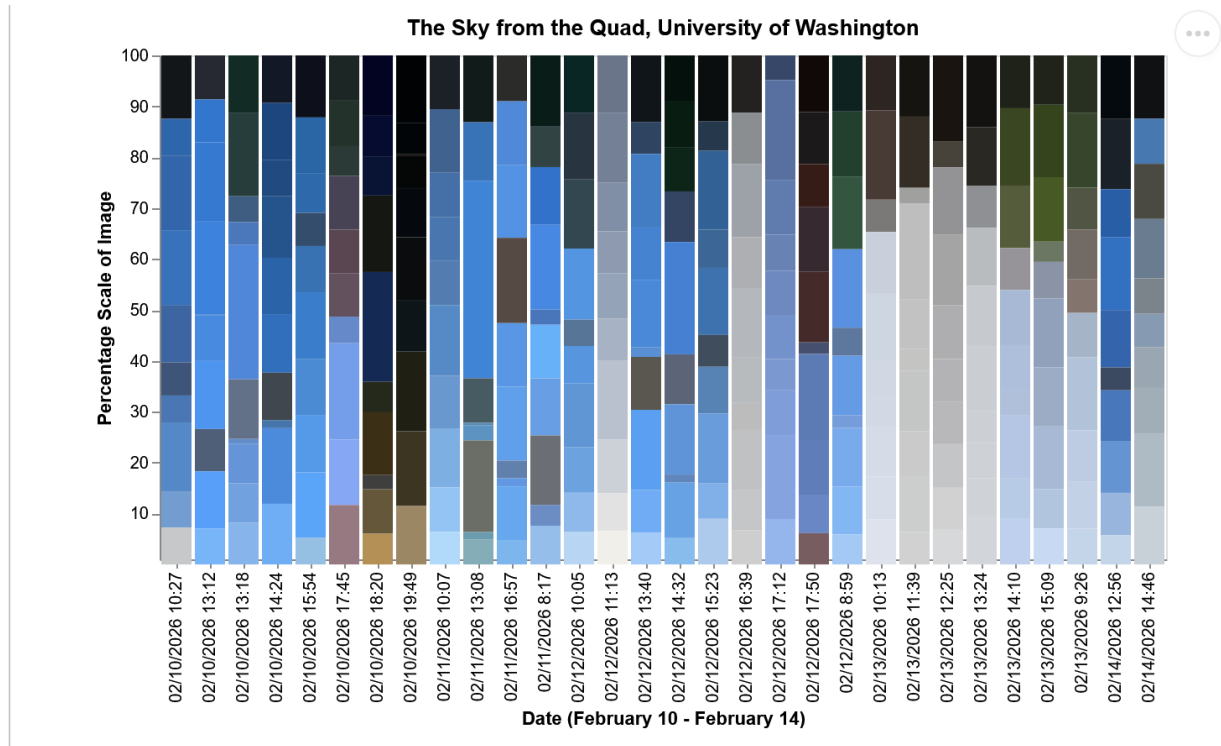
Our understanding was the sky should be the focus, so we brainstormed locations based on their view of the sky. Out of several options [1], we chose the Quad next to Smith Hall. This is where class takes place and was the most practical location for consistent access. We decided to focus our images primarily on the sky with some ground material like Smith Hall or the grass.

The instructions required at least 30 pictures between Tuesday February 10th and Monday February 16th. Most of us were not on campus during the weekend and the 16th was a holiday. The data collection dates for this project were February 10th to 14th.

We took turns snapping pictures between classes and whenever else we were available. Since our schedules changed each day, we couldn't maintain a consistent routine. We still managed to capture a range of hours. We kept at least an hour between pictures to prevent overlap and increase our range.

With our 30 images, we used the provided Colab notebook [2] to extract the 10 primary colors of each image and save them into a Comma Separated Values (.csv) data file. This data was inputted into an Altair bar chart via a second Colab notebook [3] to show the relative quantities of the colors in each image.

The Infographic



This is the infographic we produced [3]. On the x-axis are the date and time of each image (adjusted to a 24-hour clock). On the y-axis are the percentage values. The ten primary colors per image are stacked, and their relative size correlates with the percentage of the image they occupy.

The high volume of blue suggests (correctly) that most of the images were of clear skies. The end of the week saw cloudy weather, hence the images composed mainly of grey tones. The final two images of February 10th were at night. They are the only images composed of dark blue and black tones. The green and brown tones in all the images correlate with grass and the buildings in the Quad.

Reflections

The weather this week was unusually clear and sunny for winter in Seattle. Some cloud cover arrived later in the week, even then sometimes followed by clear skies. We didn't experience heavy rain during the data collection period. These factors meant the color variation was narrower than I expected.

Several time limitations affected our data collection. We couldn't follow a scheduled cadence for gathering images, as our available hours shifted from day to day. We were

generally unable to gather images at night. We were fortunate that several team members were willing to take some early morning images and a few final pictures on Saturday Feb 14th. For similar reasons we couldn't utilize the full data collection period and missed data from Sunday 15th and Monday 16th.

Some variation may stem from difference in the observer. Some of us favored a shallower angle that included a view of the Quad and Smith Hall as well as the sky. Others preferred focusing on Smith and the sky, and some images were almost entirely of the sky. Most of this variation was within the first few days, before we settled on our preferred styles.

It's noteworthy that different phones from different brands were used during data collection. In addition to our personal photography choices, our devices likely introduced some variation into the images.

One more note about the weather during that time period [4] [5]. Someone looking at this chart, would get a decent sense of the colors visible near Smith Hall for each picture. What they wouldn't get is the temperature. Despite the clear skies, most of that week was cool and chilly. My guess is we could have obtained a similar chart if this project was conducted later in the spring or in the summer.

References

1. Jotboard link: <https://www.jotboard.com/app/boards/BHsJSOenCF9IjF8EYbEy>
2. Image Processing code link:
https://colab.research.google.com/drive/1q_q_f2rl2rjwDpf7nJeWsP7atqCLE0Qm?authuser=1
3. Altair Chart code:
<https://colab.research.google.com/drive/1nt8GUQiUnNQCUakLM2M4cLa3AGMk-QCF?authuser=1#scrollTo=zdXL1XfOq-2A>
4. "Seattle, WA Monthly Weather." AccuWeather, www.accuweather.com/en/us/seattle/98104/february-weather/351409. Accessed 22 Feb. 2026.
5. "Monthly Weather Forecast for Seattle, Washington." *The Weather Channel*, The Weather Company, 22 Feb. 2026,

weather.com/weather/monthly/l/0e671a013769a86a69535aacb59b4c5fe0d4c9ea059a089d346d96d2bb64b689.